

EV Market Segmentation

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1. Machine Learning Model Used in the Project

The K-Means clustering algorithm was employed to perform market segmentation. This unsupervised learning technique is well-suited for grouping data points into clusters based on their similarity. To optimize performance, I preprocessed the data using StandardScaler for scaling numerical variables and reduced dimensionality with Principal Component Analysis (PCA). PCA ensured that the data was not only visualized effectively but also free from noise, aiding in better clustering. The Elbow Method was used to determine the optimal number of clusters, leading to an efficient and insightful segmentation.

2. Final Conclusions & Insights

After performing segmentation, the dataset was divided into three distinct clusters based on key features such as price, range, efficiency, and body style:

- **Cluster 1:** High-end, luxury vehicles characterized by higher prices, superior range, and lower efficiency. These cars cater to premium buyers who prioritize performance and brand value.
- **Cluster 2:** Budget-friendly vehicles with moderate range and efficiency, appealing to cost-conscious buyers who need practical solutions for daily commuting.
- **Cluster 3:** Mid-range vehicles with balanced features, targeting customers who seek value for money without compromising on quality.

Key insights include:

- Price and Range are strong indicators of customer preferences.
- Body style (e.g., SUV, Sedan) heavily influences segmentation, as it aligns with lifestyle needs.
- Efficiency is a differentiator, especially for environmentally conscious consumers.

3. Improvements with Additional Time and Budget

With more resources, the project could be enhanced as follows:

Datasets Collection

- **Customer Demographics:** Information like age, income, and region would provide better consumer profiling.

- **Brand Perception Metrics:** Data on customer loyalty and satisfaction with specific brands could refine segmentation.
- **Purchase History:** Insights into past buying behaviors, financing options, and frequency of upgrades.
- **Geographic Data:** Vehicle preferences across regions or countries to identify location-based trends.
- **Real-World Performance Metrics:** Data on maintenance costs and real-world efficiency to gauge long-term value.

Additional ML Models

- **Gaussian Mixture Models (GMM):** For probabilistic clustering that can handle overlapping segments.
- **Hierarchical Clustering:** To explore a tree-like structure for identifying sub-segments.
- **DBSCAN:** For identifying clusters of varying densities, especially for niche segments.
- **AutoML Platforms:** To automate the process of model selection and hyperparameter tuning.

4. Estimated Market Size

Based on industry data for the electric vehicle (EV) market, the estimated global market size for this domain is approximately **\$800 billion by 2030**, growing at a compound annual growth rate (CAGR) of 24.3%. The dataset represents a niche within this domain, focusing on vehicles with specific characteristics, which constitutes a significant subset of the market.

5. Top Variables for Optimal Segmentation

The most influential variables for creating optimal market segments in this domain are:

1. **PriceEuro:** A direct indicator of affordability and luxury.
2. **Range_Km:** Determines the vehicle's practicality for different use cases.
3. **BodyStyle:** Reflects consumer preferences based on lifestyle (e.g., family-oriented vs. individual use).
4. **Efficiency_WhKm:** Captures eco-friendliness and operating cost considerations.